1 solution of questions 4

We need to verify whether the function

$$\langle u, v\rangle = u_1v_1 + 2u_2v_2 + u_3v_3$$

defines an inner product on $\mathbb{R}^3$ and check the orthogonality of given vectors.

Step 1: Checking Inner Product Properties

To be a valid inner product, the function must satisfy the following properties:

1. **Linearity in the first argument:** Let $u, v, w \in \mathbb{R}^3$ and α be a scalar. We check:

   $$\langle \alpha u + v, w\rangle = (\alpha u_1 + v_1)w_1 + 2(\alpha u_2 + v_2)w_2 + (\alpha u_3 + v_3)w_3.$$

   Expanding, we get:

   $$\alpha(u_1w_1 + 2u_2w_2 + u_3w_3) + (v_1w_1 + 2v_2w_2 + v_3w_3).$$

   Since this equals $\alpha\langle u, w\rangle + \langle v, w\rangle$, linearity holds.

2. **Symmetry:** We check whether $\langle u, v\rangle = \langle v, u\rangle$.

   $$\langle u, v\rangle = u_1v_1 + 2u_2v_2 + u_3v_3.$$

   Since addition is commutative, we get:

   $$\langle v, u\rangle = v_1u_1 + 2v_2u_2 + v_3u_3 = \langle u, v\rangle.$$

   Hence, symmetry holds.

3. **Positive Definiteness:** We check if $\langle u, u\rangle \geq 0$ for all u and equals 0 if and only if $u = 0$:

   $$\langle u, u\rangle = u_1^2 + 2u_2^2 + u_3^2.$$

   Since $u_1^2, 2u_2^2, u_3^2$ are all non-negative and sum to zero only when $u_1 = u_2 = u_3 = 0$, positive definiteness holds.

Since all properties hold, $\langle u, v\rangle$ is a valid inner product.

Step 2: Checking Orthogonality

Vectors $u = (1, -1, 2)$ and $v = (2, 1, 0)$ are orthogonal if:

$$\langle u, v\rangle = 1(2) + 2(-1)(1) + 2(0) = 2 - 2 + 0 = 0.$$

Since $\langle u, v\rangle = 0$, the vectors are orthogonal under this inner product.

Final Answer

- The given function defines a valid inner product on $\mathbb{R}^3$. - The vectors $u = (1, -1, 2)$ and $v = (2, 1, 0)$ are orthogonal under this inner product.

2 solution of question 6

We are given the inner product on $\mathbb{R}^3$ as:

$$\langle u, v\rangle = 3u_1v_1 + 2u_2v_2 + u_3v_3.$$

To be a valid inner product, the function must satisfy:

1. **Linearity in the first argument:** Let $u, v, w \in \mathbb{R}^3$ and α be a scalar. We check:

$$\langle \alpha u + v, w\rangle = 3(\alpha u_1 + v_1)w_1 + 2(\alpha u_2 + v_2)w_2 + (\alpha u_3 + v_3)w_3.$$

Expanding, we get:

$$\alpha(3u_1w_1 + 2u_2w_2 + u_3w_3) + (3v_1w_1 + 2v_2w_2 + v_3w_3).$$

This equals $\alpha\langle u, w\rangle + \langle v, w\rangle$, so linearity holds.

2. **Symmetry:** We check whether $\langle u, v\rangle = \langle v, u\rangle$.

$$\langle u, v\rangle = 3u_1v_1 + 2u_2v_2 + u_3v_3.$$

Since addition is commutative, we get:

$$\langle v, u\rangle = 3v_1u_1 + 2v_2u_2 + v_3u_3 = \langle u, v\rangle.$$

Hence, symmetry holds.

3. **Positive Definiteness:** We check if $\langle u, u\rangle \geq 0$ for all u and equals 0 if and only if $u = 0$:

$$\langle u, u\rangle = 3u_1^2 + 2u_2^2 + u_3^2.$$

Since all squared terms are non-negative and sum to zero only when $u_1 = u_2 = u_3 = 0$, positive definiteness holds.

Thus, $\langle u, v\rangle$ is a valid inner product.

(b) Finding the Norm of $u = (1, 2, -1)$

The norm is given by:

$$\|u\| = \sqrt{\langle u, u\rangle}.$$

Computing:

$$\langle u, u\rangle = 3(1)^2 + 2(2)^2 + (-1)^2 = 3(1) + 2(4) + 1 = 3 + 8 + 1 = 12.$$

Thus,

$$\|u\| = \sqrt{12} = 2\sqrt{3}.$$

(c) Computing the Angle Between u and v

The angle θ is given by:

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\|\|v\|}.$$

First, compute $\langle u, v\rangle$:

$$\langle u, v\rangle = 3(1)(2) + 2(2)(-1) + (-1)(3) = 6 - 4 - 3 = -1.$$

Next, compute $\|v\|$:

$$\langle v, v\rangle = 3(2)^2 + 2(-1)^2 + (3)^2 = 3(4) + 2(1) + 9 = 12 + 2 + 9 = 23.$$

$$\|v\| = \sqrt{23}.$$

Now, calculate $\cos\theta$:

$$\cos\theta = \frac{-1}{(2\sqrt{3})(\sqrt{23})} = \frac{-1}{2\sqrt{69}}.$$

Thus, the angle is:

$$\theta = \cos^{-1}\left(\frac{-1}{2\sqrt{69}}\right).$$

Final Answer

- The given function defines a valid inner product on $\mathbb{R}^3$.
- The norm of $u = (1, 2, -1)$ is $2\sqrt{3}$.
- The angle between u and v is $\cos^{-1}\left(\frac{-1}{2\sqrt{69}}\right)$.

Solution of question 5

We are given the matrix A as follows:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}.$$

We will apply the Gram-Schmidt process to find an orthonormal basis for the column space of A, and then verify the orthogonality of the obtained basis vectors.

(a) **Apply the Gram-Schmidt process to find an orthonormal basis for the column space of A.**

Let the columns of A be denoted as:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

Start with v_1:

$$u_1 = v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Now normalize u_1 to get the first orthonormal vector:

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{1^2 + 1^2 + 1^2}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Next, construct u_2 by subtracting the projection of v_2 onto e_1:

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1.$$

First compute the inner product $\langle v_2, e_1 \rangle$:

$$\langle v_2, e_1 \rangle = \left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{3}}(1 + 2 + 3) = \frac{6}{\sqrt{3}} = 2\sqrt{3}.$$

Now subtract the projection:

$$u_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 2\sqrt{3} \times \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Normalize u_2 to get the second orthonormal vector:

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{(-1)^2 + 0^2 + 1^2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Now construct u_3 by subtracting the projections of v_3 onto e_1 and e_2:

$$u_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2.$$

First compute the inner product $\langle v_3, e_1 \rangle$:

$$\langle v_3, e_1 \rangle = \left\langle \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{3}}(1 + 3 + 5) = \frac{9}{\sqrt{3}} = 3\sqrt{3}.$$

Next, compute $\langle v_3, e_2 \rangle$:

$$\langle v_3, e_2 \rangle = \left\langle \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{2}}(-1 + 5) = \frac{4}{\sqrt{2}} = 2\sqrt{2}.$$

Now subtract the projections:

$$u_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - 3\sqrt{3} \times \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 2\sqrt{2} \times \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Simplifying the terms:

$$u_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since u_3 is the zero vector, it is not contributing to the orthonormal basis. Thus, the first two vectors e_1 and e_2 are sufficient.

The orthonormal basis is therefore:

$$e_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad e_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

(b) **Verify that the obtained basis vectors are orthogonal.**

To verify orthogonality, we need to check that the inner product of each pair of basis vectors is zero:

$$\langle e_1, e_2 \rangle = \left\langle \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{2}} \left(1(-1) + 1(0) + 1(1)\right).$$

Simplifying:

$$\langle e_1, e_2 \rangle = \frac{1}{\sqrt{6}}(-1 + 0 + 1) = 0.$$

Since $\langle e_1, e_2 \rangle = 0$, the vectors are orthogonal.

Solution of question 9

We are given the inner product space of continuous functions on the interval $[0, 1]$ with the inner product defined as:

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx.$$

The functions provided are:

$$f_1(x) = x, \quad f_2(x) = x^2 - \frac{1}{3}, \quad f_3(x) = x^3 - \frac{3}{5}x.$$

(a) **Compute the inner products $\langle f_1, f_2\rangle$, $\langle f_1, f_3\rangle$, and $\langle f_2, f_3\rangle$.**
To compute these inner products, we will evaluate the integrals individually:

$$\langle f_1, f_2\rangle = \int_0^1 f_1(x)f_2(x)\,dx = \int_0^1 x\left(x^2 - \frac{1}{3}\right)\,dx.$$

Expanding the integrand:

$$\langle f_1, f_2\rangle = \int_0^1 \left(x^3 - \frac{1}{3}x\right)\,dx.$$

Now compute the integrals:

$$\int_0^1 x^3\,dx = \frac{1}{4}, \quad \int_0^1 x\,dx = \frac{1}{2}.$$

Thus:

$$\langle f_1, f_2\rangle = \left(\frac{1}{4} - \frac{1}{3}\times\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Next, compute $\langle f_1, f_3\rangle$:

$$\langle f_1, f_3\rangle = \int_0^1 f_1(x)f_3(x)\,dx = \int_0^1 x\left(x^3 - \frac{3}{5}x\right)\,dx.$$

Expanding the integrand:

$$\langle f_1, f_3\rangle = \int_0^1 \left(x^4 - \frac{3}{5}x^2\right)\,dx.$$

Now compute the integrals:

$$\int_0^1 x^4\,dx = \frac{1}{5}, \quad \int_0^1 x^2\,dx = \frac{1}{3}.$$

Thus:

$$\langle f_1, f_3\rangle = \left(\frac{1}{5} - \frac{3}{5}\times\frac{1}{3}\right) = \frac{1}{5} - \frac{1}{5} = 0.$$

Finally, compute $\langle f_2, f_3 \rangle$:

$$\langle f_2, f_3 \rangle = \int_0^1 f_2(x) f_3(x)\, dx = \int_0^1 \left(x^2 - \frac{1}{3}\right)\left(x^3 - \frac{3}{5}x\right) dx.$$

Expanding the integrand:

$$\langle f_2, f_3 \rangle = \int_0^1 \left(x^5 - \frac{3}{5}x^3 - \frac{1}{3}x^3 + \frac{1}{5}x\right) dx.$$

Now compute the integrals:

$$\int_0^1 x^5\, dx = \frac{1}{6}, \quad \int_0^1 x^3\, dx = \frac{1}{4}, \quad \int_0^1 x\, dx = \frac{1}{2}.$$

Thus:

$$\langle f_2, f_3 \rangle = \left(\frac{1}{6} - \frac{3}{5} \times \frac{1}{4} - \frac{1}{3} \times \frac{1}{4} + \frac{1}{5} \times \frac{1}{2}\right).$$

Simplifying:

$$\langle f_2, f_3 \rangle = \frac{1}{6} - \frac{3}{20} - \frac{1}{12} + \frac{1}{10}.$$

Finding a common denominator (60):

$$\langle f_2, f_3 \rangle = \frac{10}{60} - \frac{9}{60} - \frac{5}{60} + \frac{6}{60} = 0.$$

(b) **Determine whether the set $\{f_1, f_2, f_3\}$ forms an orthogonal set.**

A set of functions is orthogonal if all pairwise inner products are zero. From part (a), we have:

$$\langle f_1, f_2 \rangle = \frac{1}{12}, \quad \langle f_1, f_3 \rangle = 0, \quad \langle f_2, f_3 \rangle = 0.$$

Since $\langle f_1, f_2 \rangle \neq 0$, the set $\{f_1, f_2, f_3\}$ is not orthogonal.

(c) **Apply the Gram-Schmidt process to construct an orthogonal basis for the space spanned by $\{f_1, f_2, f_3\}$.**

To apply the Gram-Schmidt process, we will orthogonalize the functions step by step.

Start with $f_1(x)$:

$$u_1 = f_1 = x.$$

Next, construct u_2 by subtracting the projection of f_2 onto u_1 from f_2:

$$u_2 = f_2 - \frac{\langle f_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1.$$

First compute the inner products:

$$\langle f_2, u_1 \rangle = \langle x^2 - \frac{1}{3}, x \rangle = \frac{1}{12}, \quad \langle u_1, u_1 \rangle = \langle x, x \rangle = \frac{1}{3}.$$

Thus:

$$u_2 = f_2 - \frac{\frac{1}{12}}{\frac{1}{3}} x = f_2 - \frac{1}{4} x.$$

So:

$$u_2(x) = x^2 - \frac{1}{3} - \frac{1}{4} x.$$

Now construct u_3 by subtracting the projections of f_3 onto both u_1 and u_2:

$$u_3 = f_3 - \frac{\langle f_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle f_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2.$$

After calculating the necessary inner products, you will find u_3 to be the orthogonalized function.